

A

GPverdict

Choose genomic prediction methods for selection — from your own cross-validation, in your browser.

Which metric can rank methods?

Checked on your predictions: only metrics that ignore order-preserving changes and respond to order-reversal represent within-environment selection.

How large must a trial be?

Genotype–environment cells needed to resolve a given accuracy gain.

Which method, and is its lead real?

Methods ranked by the within-environment rank and Pearson correlations, with the methods your trial cannot tell apart from the leaders.

What would RMSE have chosen?

The method an error-magnitude ranking picks, and an out-of-sample test of how much selection gain each rule recovers.

Predictions (CSV)

No file chosen

One row per environment, genotype and method, with columns `environment`, `genotype`, `observed`, `predicted`, `method` (also accepted: `Env`, `k`, `y`, `p`). Predictions should come from cross-validation.

Selected fraction within each environment

top 10 %



Minimum genotypes per environment

10

[Try the spring wheat example](#)

Ready. Choose a CSV file or try the example.

Your data stay on this computer: every computation runs locally in the browser (Pyodide); nothing is uploaded.

B

GPverdict report

17 methods, 109 environments, 384 genotypes; selection of the top 10% within each environment.

Verdict

Rank methods by a within-environment correlation for selecting the top 10% in each environment: `ridge_pc40_lo` ranks first by the rank correlation and `mlp` by Pearson r . Both correlations can rank methods for this decision (the rank correlation is admissible, Pearson r is protected against calibration), and in the published outcome tests neither consistently selected better material.

Choose among `ridge_pc40_lo`, `ridge_pc40_hi`, `mlp`, `ens`, `gbm` on cost or simplicity: at this trial size (1,948 genotype–environment cells per method) they lie within 0.067 of the best on at least one correlation, the smallest gap such a trial resolves (Gaussian lower bound; calibrated panels needed 0.101–0.282).

The first two methods are practically identical on the rank correlation (gap below 0.0001).

Ranking by RMSE would pick `ens` instead; RMSE and Pearson r order 25 % of method pairs in opposite directions.

Out of sample (environment split, 200 splits, outcome reliability 0.68), the method each rule picks recovers 59 % (rank correlation), 77 % (Pearson r) and 37 % (RMSE) of the selection gain attainable over an average method.